

Brutal Practice Paper B52 - Heat, Comparing Quantities, Respiration in Organisms, Algebraic Expressions

Each question has 6 to 8 options; exactly ONE is correct. Answers are scattered - there is NO pattern. Every question needs several steps - read carefully and work it out fully.

1. A clinical thermometer reads 98.6 degF for a healthy body. A patient's reading is 2.5 degF above normal in the morning, and by evening it has fallen by 1.8 degF. What is the evening reading? [\[Heat\]](#)
 - (A) 99.3 degF
 - (B) 99.6 degF
 - (C) 100.4 degF
 - (D) 101.1 degF
 - (E) 96.1 degF
 - (F) 97.9 degF
 - (G) 102.9 degF

2. In a forest survey 1500 trees were counted: 18% were sal trees and 24% were teak. How many of the trees were neither sal nor teak? [\[Comparing Quantities\]](#)
 - (A) 630
 - (B) 58
 - (C) 270
 - (D) 420
 - (E) 900
 - (F) 870
 - (G) 360

3. An adult at rest breathes about 15 times per minute. How many breaths is that over a period of 2 hours? [\[Respiration in Organisms\]](#)
 - (A) 180
 - (B) 900
 - (C) 1500
 - (D) 3600
 - (E) 30
 - (F) 1200
 - (G) 1800

4. Find the value of the expression $3x^2 - 2x + 5$ when $x = 4$. [\[Algebraic Expressions\]](#)
 - (A) 57
 - (B) 41
 - (C) 33
 - (D) 45
 - (E) 53
 - (F) 37

5. On a beach during the day the land air is at 38 degC while the air over the sea is at 26 degC. What temperature difference drives the breeze, and which way does the daytime breeze blow? [\[Heat\]](#)
- (A) 64 degC, from sea to land
 - (B) 12 degC, from land to sea
 - (C) 26 degC, from sea to land
 - (D) 38 degC, from land to sea
 - (E) 14 degC, from sea to land
 - (F) 12 degC, no breeze blows
 - (G) 12 degC, from sea to land
6. Air is about 78% nitrogen and 21% oxygen by volume. In a 2000 mL sample of air, how much more nitrogen is there than oxygen? [\[Comparing Quantities\]](#)
- (A) 1140 mL
 - (B) 1980 mL
 - (C) 990 mL
 - (D) 1560 mL
 - (E) 420 mL
 - (F) 57 mL
 - (G) 1710 mL
7. At rest a girl breathes 18 times a minute. During hard exercise her breathing rate rises by 150%. What is her breathing rate during the exercise? [\[Respiration in Organisms\]](#)
- (A) 48 breaths per minute
 - (B) 36 breaths per minute
 - (C) 40 breaths per minute
 - (D) 23 breaths per minute
 - (E) 27 breaths per minute
 - (F) 45 breaths per minute
 - (G) 150 breaths per minute
8. Simplify the expression $7a + 3b - 2a + 5b - 4$. [\[Algebraic Expressions\]](#)
- (A) $5a + 8b - 4$
 - (B) $5a - 8b - 4$
 - (C) $9a + 8b - 4$
 - (D) $5a + 8b + 4$
 - (E) $13ab - 4$
 - (F) $9a + 2b - 4$
9. Heat travels along an iron rod so that its far end rises a steady 3 degC each minute, starting from 27 degC. After how many minutes will the far end reach 60 degC? [\[Heat\]](#)
- (A) 11 minutes
 - (B) 33 minutes
 - (C) 9 minutes
 - (D) 12 minutes
 - (E) 19 minutes
 - (F) 10 minutes
 - (G) 20 minutes

- 10.** Ravi deposits Rs 4000 in a bank at 8% simple interest per year. How much interest does he earn in 2 years and 6 months? [\[Comparing Quantities\]](#)
- (A) Rs 320
 - (B) Rs 640
 - (C) Rs 1000
 - (D) Rs 960
 - (E) Rs 720
 - (F) Rs 800
- 11.** Which word equation correctly shows aerobic respiration taking place inside our cells? [\[Respiration in Organisms\]](#)
- (A) oxygen $\rightarrow$ carbon dioxide + energy
 - (B) glucose + oxygen $\rightarrow$ carbon dioxide + water + energy
 - (C) glucose $\rightarrow$ lactic acid + energy
 - (D) glucose $\rightarrow$ alcohol + carbon dioxide + energy
 - (E) carbon dioxide + water $\rightarrow$ glucose + oxygen
 - (F) glucose + carbon dioxide $\rightarrow$ oxygen + energy
 - (G) water + energy $\rightarrow$ glucose + oxygen
- 12.** A rectangle has length $(2x + 3)$ cm and breadth $(x - 1)$ cm. Write an expression for its perimeter and find the perimeter when $x = 5$. [\[Algebraic Expressions\]](#)
- (A) $6x + 4$; 34 cm
 - (B) $3x + 2$; 17 cm
 - (C) $6x + 4$; 32 cm
 - (D) $3x + 4$; 19 cm
 - (E) $6x - 4$; 26 cm
 - (F) $6x + 8$; 38 cm
- 13.** A beaker of water heated steadily has its temperature rise (in degC) given by the expression $4t + 5$ after t minutes. By how much does the temperature rise between the 2nd and the 6th minute? [\[Heat\]](#)
- (A) 21 degC
 - (B) 42 degC
 - (C) 4 degC
 - (D) 13 degC
 - (E) 29 degC
 - (F) 16 degC
- 14.** A shopkeeper buys a fan for Rs 750 and sells it at a profit of 16%. What is the selling price of the fan? [\[Comparing Quantities\]](#)
- (A) Rs 900
 - (B) Rs 810
 - (C) Rs 870
 - (D) Rs 120
 - (E) Rs 630
 - (F) Rs 766

- 15.** During very hard exercise our muscles may respire without enough oxygen, producing a substance that causes muscle cramps. That substance is [\[Respiration in Organisms\]](#)
- (A) alcohol
 - (B) water
 - (C) lactic acid
 - (D) oxygen
 - (E) ethanol and carbon dioxide
 - (F) carbon dioxide only
 - (G) glucose
- 16.** In the expression $5x^2y - 7xy^2 + 3xy$, what is the coefficient of the term xy^2 ? [\[Algebraic Expressions\]](#)
- (A) -7
 - (B) $-7x$
 - (C) 5
 - (D) $-7xy^2$
 - (E) -1
 - (F) 3
 - (G) 7
- 17.** Over five days a town recorded 31, 34, 29, 36 and 30 degC. The sixth day was 4 degC hotter than the five-day average. What was the sixth day's temperature? [\[Heat\]](#)
- (A) 40 degC
 - (B) 35 degC
 - (C) 34 degC
 - (D) 36 degC
 - (E) 28 degC
 - (F) 32 degC
- 18.** A cycle bought for Rs 2500 is later sold for Rs 2150. What is the percentage loss? [\[Comparing Quantities\]](#)
- (A) 35%
 - (B) 12%
 - (C) 8%
 - (D) 7%
 - (E) 14% gain
 - (F) 14%
 - (G) 16%
- 19.** Yeast can respire without oxygen. In this anaerobic respiration, yeast breaks down sugar to release energy and produce [\[Respiration in Organisms\]](#)
- (A) glucose and oxygen
 - (B) lactic acid and water
 - (C) alcohol and carbon dioxide
 - (D) starch and carbon dioxide
 - (E) lactic acid and oxygen
 - (F) carbon dioxide and oxygen
 - (G) only water

20. Find the value of $2a + 3b$ when $a = \frac{1}{2}$ and $b = \frac{1}{3}$. [\[Algebraic Expressions\]](#)

- (A) 5
- (B) 1.5
- (C) 1
- (D) 2
- (E) 3
- (F) $\frac{5}{6}$
- (G) 2.5

21. Water freezes at 0 degC and boils at 100 degC. A liquid sits at exactly three-fifths of the way from the freezing point to the boiling point of water on the Celsius scale. What is its temperature? [\[Heat\]](#)

- (A) 35 degC
- (B) 40 degC
- (C) 53 degC
- (D) 65 degC
- (E) 80 degC
- (F) 60 degC

22. A city's average summer temperature rose from 25 degC to 29 degC over ten years. What is the percentage increase in the Celsius reading? [\[Comparing Quantities\]](#)

- (A) 4%
- (B) 16%
- (C) 20%
- (D) 8%
- (E) 25%
- (F) 14%
- (G) 13.8%

23. A person's lungs take in about 500 mL of air with each normal breath, breathing 16 times a minute. How many litres of air pass through the lungs in 5 minutes? [\[Respiration in Organisms\]](#)

- (A) 400 L
- (B) 20 L
- (C) 80 L
- (D) 8 L
- (E) 16 L
- (F) 40 L
- (G) 4 L

24. Sita is y years old. Her father's age is 4 years more than three times her age. Write an expression for the father's age. [\[Algebraic Expressions\]](#)

- (A) $3 + 4y$
- (B) $4y + 3$
- (C) $3y - 4$
- (D) $3y + 4$
- (E) $12y$
- (F) $3(y + 4)$

25. A dark shirt absorbs 80% of the heat falling on it, while a white shirt absorbs only 30%. If 250 units of heat fall on each shirt, how many more units does the dark shirt absorb than the white one? [\[Heat\]](#)

- (A) 200 units
- (B) 125 units
- (C) 75 units
- (D) 275 units
- (E) 50 units
- (F) 160 units
- (G) 110 units

26. In a class the ratio of students who walk to school to those who cycle is 5 : 3. What percentage of these students cycle to school? [\[Comparing Quantities\]](#)

- (A) 40%
- (B) 30%
- (C) 3%
- (D) 37.5%
- (E) 62.5%
- (F) 35%
- (G) 60%

27. Suppose each gram of glucose releases about 16 units of energy during respiration. How much energy is released from 2.5 g of glucose? [\[Respiration in Organisms\]](#)

- (A) 32 units
- (B) 20 units
- (C) 18.5 units
- (D) 6.4 units
- (E) 16 units
- (F) 64 units
- (G) 40 units

28. In a row of matchstick squares, the first square needs 4 sticks and each extra square needs 3 more. Write the number of sticks for n squares, and find how many are needed for 10 squares. [\[Algebraic Expressions\]](#)

- (A) $4n$; 40 sticks
- (B) $3n + 1$; 34 sticks
- (C) $4n - 1$; 39 sticks
- (D) $3n + 1$; 31 sticks
- (E) $3n + 1$; 30 sticks
- (F) $n + 3$; 13 sticks

29. A thermos flask cuts down all three ways heat travels. Its shiny silvered walls are there mainly to reduce which way of heat transfer? [\[Heat\]](#)

- (A) convection
- (B) evaporation
- (C) conduction
- (D) radiation
- (E) melting
- (F) conduction and convection together
- (G) condensation

30. A jacket marked Rs 1200 is sold at a 25% discount, and then Rs 40 is charged for delivery. What does the customer finally pay? [\[Comparing Quantities\]](#)

- (A) Rs 300
- (B) Rs 940
- (C) Rs 920
- (D) Rs 860
- (E) Rs 1140
- (F) Rs 900

31. Insects such as cockroaches take in air through tiny openings along the sides of their body. These openings are called [\[Respiration in Organisms\]](#)

- (A) nostrils
- (B) alveoli
- (C) lungs
- (D) stomata
- (E) gills
- (F) tracheae
- (G) spiracles

32. If $x = 2$ and $y = -3$, find the value of the expression $x^2y + xy^2$. [\[Algebraic Expressions\]](#)

- (A) 18
- (B) 6
- (C) -6
- (D) 12
- (E) -12
- (F) -30
- (G) 30

33. A lab thermometer is marked from -10 degC to 110 degC, while a clinical thermometer is marked from 35 degC to 42 degC. How much wider is the lab thermometer's range than the clinical one's? [\[Heat\]](#)

- (A) 110 degC
- (B) 127 degC
- (C) 100 degC
- (D) 7 degC
- (E) 113 degC
- (F) 106 degC
- (G) 120 degC

34. 40% of a number is 56. What is 75% of the same number? [\[Comparing Quantities\]](#)

- (A) 84
- (B) 120
- (C) 112
- (D) 105
- (E) 56
- (F) 140

- 35.** Fish breathe using gills, which take in oxygen dissolved in water. Because water holds far less oxygen than air does, a fish must [\[Respiration in Organisms\]](#)
- (A) close its gills to trap oxygen
 - (B) store oxygen in its scales
 - (C) drink the water to get oxygen
 - (D) keep passing large amounts of water over its gills
 - (E) stop moving to save oxygen
 - (F) come to the surface to breathe air
- 36.** Simplify the expression $4(p + 2) - 3(p - 1)$, then find its value when $p = 6$. [\[Algebraic Expressions\]](#)
- (A) $7p + 11$; 53
 - (B) $p + 5$; 11
 - (C) $p - 11$; -5
 - (D) $p + 11$; 17
 - (E) $p + 11$; 18
 - (F) $7p + 5$; 47
- 37.** Railway tracks are laid with small gaps so steel can expand in heat. A rail expands by 3 mm for every 10 degC rise. By how much does it expand when the temperature climbs from 20 degC to 60 degC? [\[Heat\]](#)
- (A) 18 mm
 - (B) 24 mm
 - (C) 3 mm
 - (D) 15 mm
 - (E) 6 mm
 - (F) 9 mm
 - (G) 12 mm
- 38.** A water tank is 80% full. Then 25% of the water that is in the tank is used. What percent of the full tank is left now? [\[Comparing Quantities\]](#)
- (A) 60%
 - (B) 20%
 - (C) 75%
 - (D) 65%
 - (E) 45%
 - (F) 40%
 - (G) 55%
- 39.** A newborn baby breathes about 40 times a minute, while an adult breathes about 16 times a minute. How many more breaths per minute does the newborn take, and that extra is what percentage of the adult's rate? [\[Respiration in Organisms\]](#)
- (A) 24 breaths, 150%
 - (B) 24 breaths, 40%
 - (C) 56 breaths, 150%
 - (D) 16 breaths, 150%
 - (E) 24 breaths, 60%
 - (F) 24 breaths, 250%
 - (G) 40 breaths, 100%

40. A square has side $(x + 2)$ cm. Write an expression for its area and find the area when $x = 3$. [\[Algebraic Expressions\]](#)

- (A) $x^2 + 4$; 13 cm²
- (B) $x^2 + 4x + 4$; 23 cm²
- (C) $x^2 + 2x + 4$; 19 cm²
- (D) $x^2 + 4x + 2$; 23 cm²
- (E) $4x + 8$; 20 cm²
- (F) $2x + 4$; 10 cm²
- (G) $x^2 + 4x + 4$; 25 cm²

41. A cup of tea cools from 80 degC to 60 degC in ten minutes. What percentage of its starting Celsius reading has it lost? [\[Heat\]](#)

- (A) 75%
- (B) 33.3%
- (C) 12.5%
- (D) 25%
- (E) 40%
- (F) 30%
- (G) 20%

42. A plant loses 6 mL of water through transpiration in 4 hours. At the same steady rate, how much water will it lose in a full day? [\[Comparing Quantities\]](#)

- (A) 18 mL
- (B) 36 mL
- (C) 9 mL
- (D) 12 mL
- (E) 24 mL
- (F) 144 mL
- (G) 48 mL

43. Plants take in and give out gases for respiration mainly through tiny pores found on the surface of their leaves. These pores are called [\[Respiration in Organisms\]](#)

- (A) spiracles
- (B) root hairs
- (C) tracheae
- (D) lenticels
- (E) alveoli
- (F) stomata
- (G) the cuticle

44. How many terms are there in the expression $3x^2y - 5x + 7 - 2xy$? [\[Algebraic Expressions\]](#)

- (A) 1
- (B) 3
- (C) 4
- (D) 5
- (E) 2
- (F) 6
- (G) 7

45. On a winter morning the temperature was -3°C at dawn and rose to 17°C by noon. What temperature is exactly halfway between these two readings? [\[Heat\]](#)

- (A) 12°C
- (B) 10°C
- (C) -3°C
- (D) 7°C
- (E) 14°C
- (F) 8°C
- (G) 20°C

46. Rs 5400 is shared between two friends in the ratio 4 : 5. How much more money does the second friend get than the first? [\[Comparing Quantities\]](#)

- (A) Rs 900
- (B) Rs 100
- (C) Rs 2400
- (D) Rs 1200
- (E) Rs 540
- (F) Rs 3000
- (G) Rs 600

47. Plants growing in waterlogged soil often die. The best reason is that [\[Respiration in Organisms\]](#)

- (A) roots respire faster in water and use up food
- (B) water freezes around the roots
- (C) the soil becomes too acidic for roots
- (D) roots need carbon dioxide from soil air, which water blocks
- (E) roots stop absorbing sunlight
- (F) water stops the leaves from making food
- (G) their roots cannot get enough oxygen for respiration

48. Add the two expressions $3x^2 - 2x + 1$ and $-x^2 + 5x - 4$. [\[Algebraic Expressions\]](#)

- (A) $2x^2 + 7x - 3$
- (B) $2x^2 + 3x + 5$
- (C) $2x^2 + 3x - 4$
- (D) $4x^2 + 3x - 3$
- (E) $2x^2 - 3x - 3$
- (F) $2x^2 + 3x - 5$
- (G) $2x^2 + 3x - 3$

49. Four thermometers in a greenhouse read 22, 25, 24 and 29°C . A fifth thermometer is added, and the average of all five becomes 27°C . What did the fifth thermometer read? [\[Heat\]](#)

- (A) 25°C
- (B) 31°C
- (C) 33°C
- (D) 27°C
- (E) 35°C
- (F) 30°C
- (G) 40°C

50. After a price rise of 20%, a gas cylinder now costs Rs 960. What was its price before the rise? [\[Comparing](#)

[Quantities\]](#)

- (A) Rs 800
- (B) Rs 1152
- (C) Rs 780
- (D) Rs 768
- (E) Rs 940
- (F) Rs 760
- (G) Rs 820

51. When we breathe out onto limewater, it turns milky. This shows that the air we breathe out is rich in

[\[Respiration in Organisms\]](#)

- (A) carbon dioxide
- (B) water vapour only
- (C) nitrogen
- (D) hydrogen
- (E) lactic acid
- (F) oxygen
- (G) alcohol

52. Subtract the expression $2a - 3b$ from $5a + b$. [\[Algebraic Expressions\]](#)

- (A) $3a - 2b$
- (B) $-3a + 4b$
- (C) $3a + 4b$
- (D) $3a + 2b$
- (E) $7a - 2b$
- (F) $3a - 4b$
- (G) $7a + 4b$

53. Two identical pots of water, one black and one shiny, are left in the Sun. In the same time the black pot's water rises 18 degC and the shiny pot's rises 6 degC. What is the ratio of heat absorbed by the black pot to the shiny pot? [\[Heat\]](#)

- (A) 12 : 1
- (B) 18 : 1
- (C) 6 : 18
- (D) 2 : 1
- (E) 9 : 1
- (F) 1 : 3
- (G) 3 : 1

54. In a test of 80 marks Meena scored 60. What percentage did she score, and how many more marks did she need to reach 90%? [\[Comparing Quantities\]](#)

- (A) 60%, 12 more marks
- (B) 70%, 12 more marks
- (C) 75%, 30 more marks
- (D) 75%, 10 more marks
- (E) 80%, 8 more marks
- (F) 75%, 12 more marks
- (G) 75%, 20 more marks

- 55.** At rest the body uses about 250 mL of oxygen per minute. During a 30-minute jog this rate triples. How much oxygen is used during the whole jog? [\[Respiration in Organisms\]](#)
- (A) 11.25 L
 - (B) 750 L
 - (C) 22.5 L
 - (D) 45 L
 - (E) 15 L
 - (F) 225 L
 - (G) 7.5 L
- 56.** Find the value of the expression $-2x^2$ when $x = -3$. [\[Algebraic Expressions\]](#)
- (A) 6
 - (B) -36
 - (C) -12
 - (D) 36
 - (E) 18
 - (F) 12
 - (G) -18
- 57.** A lab thermometer's smallest division is 0.5 degC. The top of the mercury thread sits two whole divisions above the 36 degC mark. What temperature does it read? [\[Heat\]](#)
- (A) 36.5 degC
 - (B) 38 degC
 - (C) 36.05 degC
 - (D) 37 degC
 - (E) 37.5 degC
 - (F) 36.2 degC
 - (G) 40 degC
- 58.** A 50 g sample of milk contains 4% fat and 5% protein by mass. What is the total mass of fat and protein together in the sample? [\[Comparing Quantities\]](#)
- (A) 9 g
 - (B) 2 g
 - (C) 5 g
 - (D) 4 g
 - (E) 0.9 g
 - (F) 4.5 g
 - (G) 2.5 g
- 59.** Inhaled air contains about 0.04% carbon dioxide, while exhaled air contains about 4%. About how many times more carbon dioxide is in exhaled air than in inhaled air? [\[Respiration in Organisms\]](#)
- (A) 1000 times
 - (B) 25 times
 - (C) 100 times
 - (D) 40 times
 - (E) 50 times
 - (F) 10 times
 - (G) 4 times

60. The cost of buying 3 notebooks at Rs x each and 2 pens at Rs y each is best written as which expression? [\[Algebraic Expressions\]](#)

- (A) $3y + 2x$ apart
- (B) $3x + 2y$
- (C) $6xy$
- (D) $5(x + y)$
- (E) $3 + 2 + x + y$
- (F) $5xy$

61. In the same sunlight sand heats up about five times as fast as water. If the water in a pond warms by 2 degC in an hour, by about how much does nearby sand warm in the same hour? [\[Heat\]](#)

- (A) 12 degC
- (B) 2.5 degC
- (C) 20 degC
- (D) 7 degC
- (E) 10 degC
- (F) 5 degC
- (G) 0.4 degC

62. Two square plots of land have sides in the ratio 2 : 3. What is the ratio of their areas? [\[Comparing Quantities\]](#)

- (A) 4 : 9
- (B) 6 : 9
- (C) 16 : 81
- (D) 8 : 27
- (E) 4 : 6
- (F) 2 : 3
- (G) 9 : 4

63. When we breathe in, the muscular sheet below the lungs called the diaphragm [\[Respiration in Organisms\]](#)

- (A) stays still while only the ribs move down
- (B) domes upward, so the chest cavity gets smaller
- (C) squeezes the lungs to push air in
- (D) flattens and moves down, so the chest cavity gets smaller
- (E) domes upward, so the chest cavity gets larger
- (F) flattens and moves down, so the chest cavity gets larger
- (G) relaxes so air is forced out

64. An equilateral triangle has each side equal to $(2x - 1)$ cm, and its perimeter is 33 cm. Find the value of

X. [\[Algebraic Expressions\]](#)

- (A) 5.5
- (B) 6
- (C) 5
- (D) 3
- (E) 17
- (F) 6.5

65. A room heater raises a room from 14 degC to 26 degC. The family then wants to settle at a comfortable 22 degC. By how many degrees must the temperature be reduced from 26 degC, and what fraction of the total rise does that reduction undo? [\[Heat\]](#)

- (A) 4 degC, one-third
- (B) 4 degC, one-half
- (C) 12 degC, one-third
- (D) 6 degC, one-third
- (E) 8 degC, two-thirds
- (F) 4 degC, one-sixth
- (G) 4 degC, one-fourth

66. A trader buys 12 pens for Rs 96 and sells them at Rs 10 each. What is his percentage profit? [\[Comparing Quantities\]](#)

- (A) 25%
- (B) 10%
- (C) 20%
- (D) 2%
- (E) 16.7%
- (F) 25% loss
- (G) 12.5%

67. Earthworms breathe through their moist skin. This is why earthworms crawl out of the soil when it is flooded by heavy rain, because [\[Respiration in Organisms\]](#)

- (A) sunlight after rain attracts them out
- (B) water fills the soil spaces and they cannot get enough oxygen through their skin
- (C) they breathe better in open air using lungs
- (D) they come out to drink the rainwater
- (E) rain washes away their food underground
- (F) the soil becomes too warm for them

68. Find the value of the expression $(a^2 - b^2) / (a - b)$ when $a = 7$ and $b = 3$. [\[Algebraic Expressions\]](#)

- (A) 10
- (B) 58
- (C) 49
- (D) 4
- (E) 40
- (F) 13

69. A workshop compares wires. Copper conducts heat well and costs Rs 90 per metre; aluminium conducts a little less and costs Rs 60 per metre. The workshop buys 6 m of copper and 4 m of aluminium. What is the total cost? [\[Heat\]](#)

- (A) Rs 900
- (B) Rs 240
- (C) Rs 600
- (D) Rs 1500
- (E) Rs 780
- (F) Rs 720

70. A toy is priced at Rs 500. A 10% discount is given first, and then an 8% tax is added on the discounted price. What is the final price? [\[Comparing Quantities\]](#)
- (A) Rs 510
 - (B) Rs 468
 - (C) Rs 490
 - (D) Rs 540
 - (E) Rs 486
 - (F) Rs 450
 - (G) Rs 500
71. If a person breathes 20 times every minute, how many breaths do they take in one full day? [\[Respiration in Organisms\]](#)
- (A) 2880
 - (B) 1200
 - (C) 14400
 - (D) 480
 - (E) 28800
 - (F) 20000
 - (G) 57600
72. Simplify the expression $5xy - 3yz + 2xy + 4yz - xy$. [\[Algebraic Expressions\]](#)
- (A) $6xy - yz$
 - (B) $8xy + yz$
 - (C) $6xy + 7yz$
 - (D) $6xy + yz$
 - (E) $6xy - 7yz$
 - (F) $7xy + yz$
 - (G) $5xy + yz$
73. On a cold morning a metal chair feels colder than a wooden chair, even though both are at the same temperature. The best reason is that [\[Heat\]](#)
- (A) the metal is actually colder than the wood
 - (B) wood produces heat of its own
 - (C) the metal is at a much lower temperature
 - (D) metal conducts heat away from your hand faster than wood does
 - (E) air around metal is always colder
 - (F) wood absorbs your body heat faster than metal
 - (G) metal reflects cold onto your skin
74. Which of these is the largest value: $\frac{3}{5}$, 0.58, 62%, or $\frac{11}{20}$? [\[Comparing Quantities\]](#)
- (A) $\frac{3}{5}$
 - (B) 55%
 - (C) they are all equal
 - (D) 0.6
 - (E) 0.58
 - (F) 62%

75. Three resting people breathe at 16, 20 and 24 times per minute. What is their average breathing rate?

[Respiration in Organisms]

- (A) 22 breaths per minute
- (B) 18 breaths per minute
- (C) 60 breaths per minute
- (D) 24 breaths per minute
- (E) 21 breaths per minute
- (F) 20 breaths per minute
- (G) 19 breaths per minute

76. The energy released by a reaction is given by the expression $E = 5m + 3$ units, where m is the mass in grams. Find E when $m = 8$ g, and the increase in E as m rises from 8 g to 12 g. [Algebraic Expressions]

- (A) 43 units; increase 20 units
- (B) 48 units; increase 20 units
- (C) 63 units; increase 20 units
- (D) 40 units; increase 20 units
- (E) 43 units; increase 63 units
- (F) 43 units; increase 24 units
- (G) 43 units; increase 16 units

77. A stove takes 8 minutes to bring 1 litre of water to a boil. Assuming the time needed is proportional to the amount of water, about how long will it take to boil 2.5 litres on the same flame? [Heat]

- (A) 10 minutes
- (B) 8 minutes
- (C) 20 minutes
- (D) 16 minutes
- (E) 12.5 minutes
- (F) 25 minutes
- (G) 18 minutes

78. A wildlife park has 250 deer. Over one year their number increases by 12%. How many deer are there at the end of the year? [Comparing Quantities]

- (A) 278
- (B) 262
- (C) 272
- (D) 280
- (E) 300
- (F) 220
- (G) 30

79. Anaerobic respiration releases much less energy than aerobic respiration from the same amount of glucose. The reason is that [Respiration in Organisms]

- (A) water blocks the energy release
- (B) the energy escapes as light
- (C) oxygen destroys most of the energy
- (D) anaerobic respiration happens too quickly to release energy
- (E) glucose is not used at all without oxygen
- (F) anaerobic respiration uses up energy instead of making it
- (G) without oxygen, glucose is only partly broken down

80. The n th term of a sequence is given by $4n - 1$. Which term of this sequence is equal to 47? [\[Algebraic Expressions\]](#)

- (A) 11th term
- (B) 47th term
- (C) 12th term
- (D) 23rd term
- (E) 6th term
- (F) 13th term

81. At night the land cools faster than the sea, so the breeze reverses. If at night the land air is 19 degC and the sea air is 25 degC, which way does the breeze blow and what is the temperature difference? [\[Heat\]](#)

- (A) from sea to land, 25 degC
- (B) from land to sea, 16 degC
- (C) from land to sea, 6 degC
- (D) from sea to land, 19 degC
- (E) from land to sea, 44 degC
- (F) from land to sea, 4 degC
- (G) from sea to land, 6 degC

82. A pond holds 4000 litres. On the first day 15% of the water evaporates, and on the second day 10% of what remains evaporates. How much water is left after the two days? [\[Comparing Quantities\]](#)

- (A) 940 L
- (B) 3000 L
- (C) 3060 L
- (D) 2960 L
- (E) 3500 L
- (F) 3100 L

83. A cell releases about 38 units of energy from one glucose molecule when it respire aerobically, but only 2 units when it respire anaerobically. The anaerobic amount is what fraction of the aerobic amount?

[\[Respiration in Organisms\]](#)

- (A) $\frac{1}{38}$
- (B) $\frac{1}{2}$
- (C) $\frac{19}{1}$
- (D) $\frac{1}{19}$
- (E) $\frac{1}{18}$
- (F) $\frac{1}{36}$

84. Multiply $3x$ by the expression $(2x - 5)$. [\[Algebraic Expressions\]](#)

- (A) $9x - 5x$
- (B) $5x^2 - 15x$
- (C) $6x^2 - 15$
- (D) $6x - 15$
- (E) $6x^2 + 15x$
- (F) $6x^2 - 15x$

85. A patient's temperature is read three times during a fever: 39.2 degC, 38.5 degC and 37.8 degC. By how much did the temperature fall in total from the first reading to the last? [\[Heat\]](#)

- (A) 1.0 degC
- (B) 0.6 degC
- (C) 1.4 degC
- (D) 2.0 degC
- (E) 0.7 degC
- (F) 2.1 degC
- (G) 1.7 degC

86. A sum of Rs 6000 earns Rs 720 in simple interest over 2 years. What is the yearly rate of interest?

[\[Comparing Quantities\]](#)

- (A) 12%
- (B) 3%
- (C) 5%
- (D) 6%
- (E) 6.5%
- (F) 4%
- (G) 8%

87. The breathed-in air finally reaches tiny balloon-like air sacs in the lungs, where oxygen passes into the blood. These air sacs are called [\[Respiration in Organisms\]](#)

- (A) stomata
- (B) bronchi
- (C) the diaphragm
- (D) gills
- (E) spiracles
- (F) alveoli

88. For the expression $P = 2x + 3$, find P when $x = 4$ and when $x = 9$, and state the difference between the two values. [\[Algebraic Expressions\]](#)

- (A) 14, 21, difference 7
- (B) 8, 18, difference 10
- (C) 10, 21, difference 11
- (D) 11, 21, difference 5
- (E) 11, 21, difference 10
- (F) 11, 21, difference 32
- (G) 11, 24, difference 13

89. Two metal blocks are heated. Block P rises from 20 degC to 50 degC and block Q rises from 25 degC to 40 degC in the same time with the same heater. The rise in P is what percent of the rise in Q? [\[Heat\]](#)

- (A) 200%
- (B) 120%
- (C) 250%
- (D) 150%
- (E) 60%
- (F) 100%

- 90.** A fertiliser mixes nitrogen, phosphorus and potash in the ratio 3 : 2 : 5. In a 40 kg bag of this fertiliser, how much potash is there? [\[Comparing Quantities\]](#)
- (A) 8 kg
 - (B) 12 kg
 - (C) 16 kg
 - (D) 20 kg
 - (E) 4 kg
 - (F) 25 kg
 - (G) 10 kg
- 91.** A swimmer holds her breath underwater for 45 seconds. If she normally breathes 12 times a minute, how many breaths does she 'miss' while holding her breath? [\[Respiration in Organisms\]](#)
- (A) 18 breaths
 - (B) 3 breaths
 - (C) 9 breaths
 - (D) 45 breaths
 - (E) 12 breaths
 - (F) 6 breaths
 - (G) 15 breaths
- 92.** Combine the like terms $(\frac{1}{2})x + (\frac{1}{3})x$ into a single term. [\[Algebraic Expressions\]](#)
- (A) $(\frac{2}{5})x$
 - (B) x
 - (C) $(\frac{2}{6})x$
 - (D) $(\frac{1}{6})x$
 - (E) $(\frac{5}{6})x$
 - (F) $(\frac{5}{6})x^2$
- 93.** A child wears a thick woollen sweater on a cold day and feels warmer. The main reason a woollen sweater works is that [\[Heat\]](#)
- (A) wool absorbs the cold from outside
 - (B) wool reflects body heat like a mirror
 - (C) air is a good conductor that carries heat inward
 - (D) wool stops air from ever touching the body
 - (E) wool makes its own heat to warm us
 - (F) wool traps air, and trapped air is a poor conductor that slows heat loss from the body
- 94.** A phone battery is 60% charged. After some use it drops to 45% charged. By what percent of the full battery did the charge fall? [\[Comparing Quantities\]](#)
- (A) 45%
 - (B) 15%
 - (C) 12%
 - (D) 30%
 - (E) 33.3%
 - (F) 20%

- 95.** During a fever a patient's breathing rate rises from 16 to 22 breaths per minute. What is the percentage increase in the breathing rate? [\[Respiration in Organisms\]](#)
- (A) 37.5%
 - (B) 27.3%
 - (C) 60%
 - (D) 12.5%
 - (E) 6%
 - (F) 37.5% decrease
- 96.** A number is increased by 7 and the result is then doubled to give 30. Writing the number as n , the situation is $2(n + 7) = 30$. Find the number n . [\[Algebraic Expressions\]](#)
- (A) 8
 - (B) 4
 - (C) 23
 - (D) 11
 - (E) 15
 - (F) 7
 - (G) 16
- 97.** A kettle's water temperature is read as 64 degC, but the thermometer always reads 2 degC too high (it has a zero error). The true temperature must rise by how many more degrees to truly reach boiling point, 100 degC? [\[Heat\]](#)
- (A) 38 degC
 - (B) 34 degC
 - (C) 2 degC
 - (D) 36 degC
 - (E) 100 degC
 - (F) 40 degC
 - (G) 62 degC
- 98.** Pack A holds 750 g of rice for Rs 60. Pack B holds 1 kg of rice for Rs 76. Which pack is cheaper per kilogram, and by how much? [\[Comparing Quantities\]](#)
- (A) Pack A, by Rs 6 per kg
 - (B) Pack B, by Rs 4 per kg
 - (C) Pack A, by Rs 16 per kg
 - (D) Pack B, by Rs 2 per kg
 - (E) Pack B, by Rs 16 per kg
 - (F) Pack A, by Rs 4 per kg
 - (G) both cost the same per kg
- 99.** Which of these animals breathes using gills throughout its whole life? [\[Respiration in Organisms\]](#)
- (A) frog
 - (B) whale
 - (C) cockroach
 - (D) earthworm
 - (E) sparrow
 - (F) fish
 - (G) human

100. Which expression is equal to $2(3x - 4) + 5$? [\[Algebraic Expressions\]](#)

- (A) $5x - 3$
- (B) $6x + 3$
- (C) $6x - 8$
- (D) $3x - 3$
- (E) $6x - 9$
- (F) $6x + 1$
- (G) $6x - 3$